Why Fair-Queueing Beats the Fick Principle: An Empirical Refutation of Homeostatic Network Control in Discrete Packet Networks

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2026

Status: focused negative-result paper (paper B of the split; paper A is paper-layering-testbed.md). The layering testbed is mentioned here only as a methodological influence; the layering results are in paper A.

Abstract

Biologically inspired control — the Fick principle (OEF), homeostatic set-points, and multidimensional kinematic baselining (MKB) — is often proposed as a way to stabilise congestion in heterogeneous backhaul. This paper tests it. Over M3–M6, on a 100-node three-medium SimPy fabric and 30 paired seeds, we compare static per-flow fair-queueing with the Fick-principle OEF actuator, the 4-D MKB trigger, an EWMA-smoothed trigger with Random Early Redirection, and a Mock Oracle stack compiler. The negative result is consistent: in this simulated testbed, under the stipulated discrete packet model, static per-flow isolation outperformed continuous bio-inspired control. We state the result as a boundary on these controllers and workloads, not a general claim about bio-inspired control, and we restate each verdict as a break-even threshold so it holds under any stipulation (see break-even-analysis.md).

1. Introduction

The motivating idea is that the Fick principle — `CMRO2 = CBF · OEF · Ca` — maps onto network metabolism: when delivered throughput (CBF) falls, an extraction fraction (OEF) should rise to protect legitimate traffic (CMRO2). M4 and M5 test this idea directly, alongside a multidimensional kinematic trigger that watches the geometry of a 4-D state vector `[CBF, OEF, Queue, Drop]`.

The testbed is the SimPy fabric of simulation/layering/ and simulation/surgical/: a 100-node mesh of fiber, microwave and satellite links under a 10 s DDoS shockwave at t = 30 s. The control-plane edit (isolation, RER) is shared with the static baseline, so the arms differ only in their trigger.

2. Controllers under test

Arm	Trigger	Actuator	Source
`static_isolation`	none (always on)	per-flow fair-queueing	M3 confound
`m4_oef`	CBF drop vs EMA	OEF weight on the bulk class	M4
`m4_raw_mkb`	4-D vector residual	surgical isolation	M4

Arm	Trigger	Actuator	Source
`m5_ewma_mkb`	EWMA of v and a + RER	surgical isolation + redirect	M5
`llm_compiler`	mocked Oracle + P1–P5 guardrails	stack ordering `C→E→F`	M5

3. What the evidence shows

- M3: the static per-flow isolation baseline delivers 86.5 % legitimate goodput under the shockwave — the number every dynamic arm must beat.
- M4: the Fick principle is refuted. The OEF actuator cannot beat static isolation (0.8465 vs 0.8648, paired $p = 0.53$): a DDoS flood raises total delivered throughput, so the CBF-drop premise that drives OEF never fires. Running OEF and MKB together is worse (goodput 0.48).
- M5: the EWMA-smoothed trigger plus RER is refuted for its peak-burst claim (the synchronous surgical edit, not the trigger, sets the peak) and is refuted on energy: it issues 7.5 % more edits than raw MKB.
- M6: the energy audit makes the boundary explicit. `static_isolation` has zero runtime churn energy and is the cheapest arm (918.2 J); the Mock Oracle compiler's EROI is $\approx 3{\cdot}10^{-6}$ (see §4). The verdicts are restated as break-even thresholds in `break-even-analysis.md`: H11 holds for any churn cost above ≈ 0.19 J/edit, and H9 is independent of the constant.

4. Stipulation dependence (why "refuted" is too strong for H10)

The H10 verdict is stipulation-dependent, not absolute. With `EROI = (Δwire · 1e-7) / (20 · N_calls)`, the compiler is viable if and only if

`E_llm_breakeven < E_tx_saved i.e. Δwire > 600 MB per 3-call transition.`

At the stipulated 20 J/call and 10^{-7} J/B, the modelled flow saves ≈ 2 KB — about five orders of magnitude short. We therefore report H10 as "stipulation-dependent" with the break-even above, not as an unconditional refutation.

5. What this does and does not show

- Shows: for this fabric, these controllers and these loads, continuous bio-inspired compensation did not beat static isolation, and lost once churn energy was counted.
- Does not show: that bio-inspired control cannot work in general. The static baseline in SimPy does not price the per-flow WFQ state maintenance that real hardware must pay (per-flow queues, hashing, decrementing timers), so the real-hardware margin is smaller than the simulated one. The result bounds this model, not the paradigm.
- The Fick analogy remains a useful design intuition; the empirical test did not confirm the mechanism under a flood.

6. Methodological influence

The survival/commutation discipline used to bound the claims here — freeze the prediction, name a cause gate, report INCONCLUSIVE rather than refuted when the setup decides the outcome — was developed in the layering testbed, paper A (`paper-layering-testbed.md`).

Cross-references

- `m6-energy-paper.md` — the energy audit in full.
- `break-even-analysis.md` — break-even thresholds.

- `../results/m4_summary.md` — M4 verdicts.
- `../results/m5/m5_summary.md` — M5 verdicts.
- `paper-layering-testbed.md` — paper A (layering).